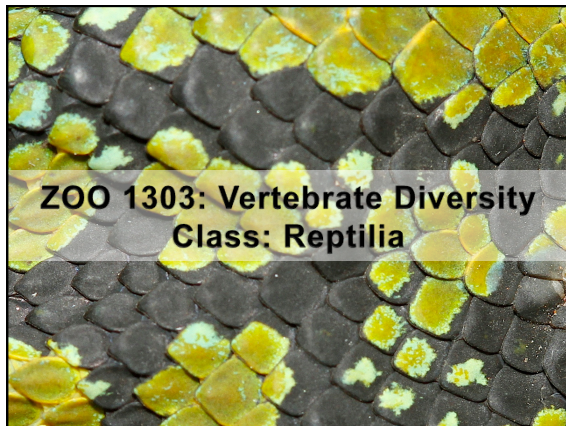




Class: Reptilia

Domain: Eukarya/ Eukaryota
Kingdom: Animalia
Phylum: Chordata
Subphylum: Vertebrata
Class: Reptilia

Class: Reptilia

Characteristics of the members of the phylum chordata

- Bilateral symmetry
- Segmented body; three germ layers
- Well developed coelom
- Notochord present at some stage in the life cycle
- Single, dorsal, tubular nerve cord
- Pharyngeal pouches present at some stage in the life cycle
- Postanal tail, projecting beyond the anus at some stage
- Segmented muscles in an unsegmented trunk
- Ventral heart, with dorsal and ventral blood vessels; closed blood system
- Complete digestive system
- Cartilaginous or bony endoskeleton

Class: Reptilia

Characteristics of the members of the subphylum vertebrata

- Notochord, dorsal nerve cord, pharyngeal pouches, and postanal tail—all present at some stage of the life cycle
- Integument- two layers: epidermis (stratified epithelium) and dermis (connective tissue)
- Cartilaginous or bony endoskeleton
- Many muscles attached to endoskeleton
- Well developed coelom (filled with visceral organs)
- Ventral heart, with dorsal and ventral blood vessels; closed blood system

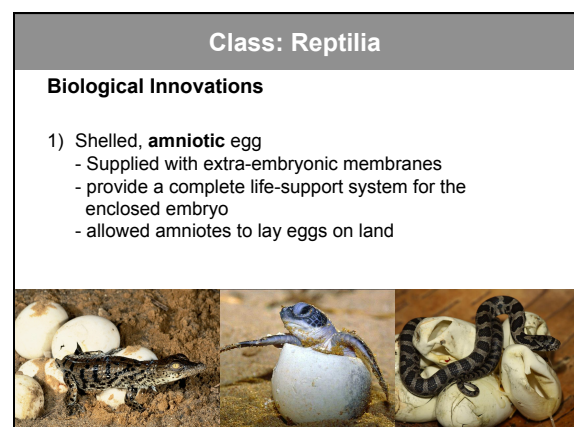
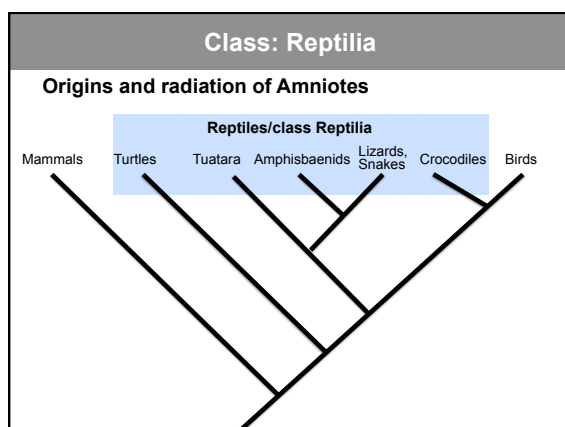
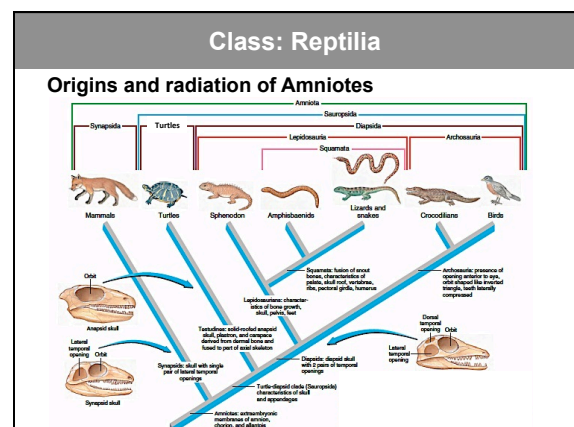
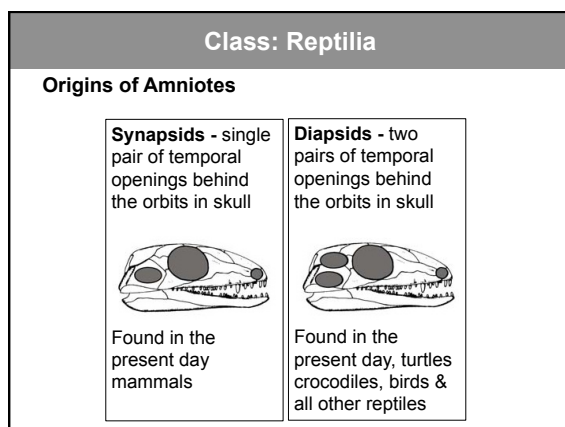
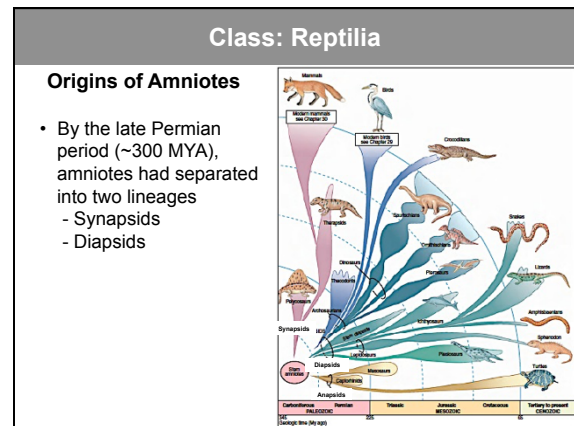
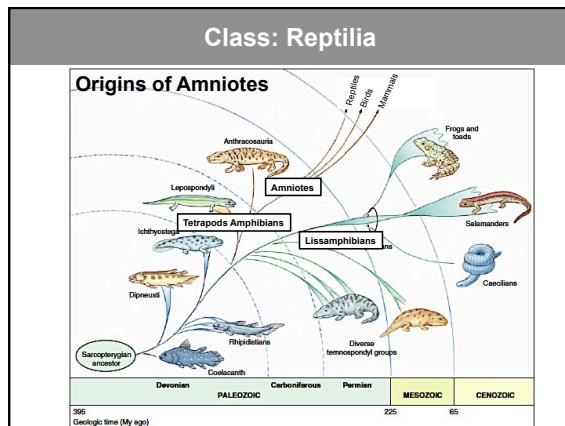
Class: Reptilia

Characteristics of the members of the subphylum vertebrata

- Complete digestive system
- Excretory system – paired kidneys
- Highly differentiated brain; 10 or 12 pairs of cranial nerves
- Endocrine system
- Separate sexes
- Body plan consisting typically of head, trunk, and postanal tail

Class: Reptilia

- Commonly called reptiles
- Reptiles – amniotic tetrapod vertebrates
- Earliest amniotes evolved in the Carboniferous period (360 - 300 Mya) from tetrapod amphibians



Class: Reptilia

Biological Innovations

- 2) A **tough, dry, heavily keratinized skin** - provides protection against desiccation and injury

Scales in reptiles and feathers in birds arise as epidermal elevations overlying a nourishing dermal layer



Class: Reptilia

Biological Innovations

- 3) Larger and stronger jaw muscles - permit **powerful jaw closure**

Temporal openings in the diapsid skull provide space for bulging temporal muscles



Class: Reptilia

Biological Innovations

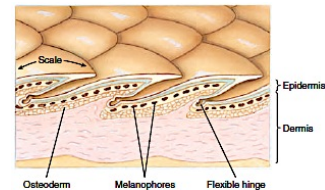
- 4) Internal fertilization
Sperm introduced directly into the female reproductive tract with a copulatory organ
- 5) Effective **adaptations for water conservation** -
Kidney that excretes nitrogenous wastes as uric acid
- allowed reptiles (and birds) to reduce water loss and occupy many terrestrial habitats

Class: Reptilia

Characteristics of Reptiles

Skin

- Tough, dry, scaly skin- offers protection against desiccation and physical injury
- Two layers in skin
 - Epidermis
 - Dermis
- Epidermis, shed periodically
- **Scales** are formed of keratin

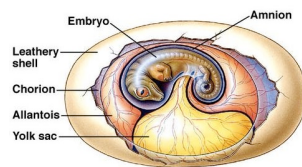


Class: Reptilia

Characteristics of Reptiles

Reproduction

- Shelled (**amniotic**) egg - contains food and protective membranes to support embryonic development on land
- **Amnion**- encloses the embryo in a fluid-filled cavity
- **Allantois** - chamber for the storage of nitrogenous wastes, respiratory surface
- **Chorion** - O₂ and CO₂ freely pass through
- Outermost membrane - porous or leathery shell
- Yolk sac - provides nutrients for the developing embryo



Class: Reptilia

Characteristics of Reptiles

Reproduction

- Copulatory organ for internal fertilization
- Sperm must reach the egg before the egg is enclosed in membranes
- Sperm from the paired testes are carried by the vasa deferentia to the copulatory organ
- Female system consists of paired ovaries and oviducts

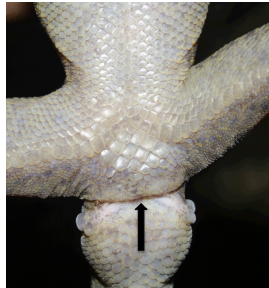


Class: Reptilia

Characteristics of Reptiles

Reproduction

- Cloaca: common opening of digestive, excretory and reproductive systems



Class: Reptilia

Characteristics of Reptiles

Feeding

- Jaws are efficiently designed for applying crushing or gripping force to prey
- Larger and longer jaw muscles
- Mainly carnivores
- Some lizards are omnivores or herbivores

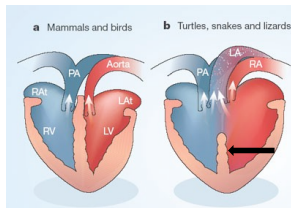


Class: Reptilia

Characteristics of Reptiles

Circulatory system

- Comparatively efficient circulatory system and higher blood pressure
- Three chambers in heart –Ventricle is incompletely separated

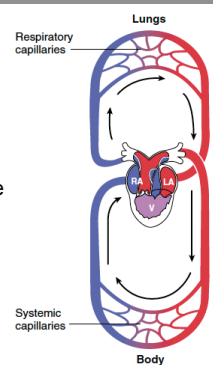


Class: Reptilia

Characteristics of Reptiles

Circulatory system

- Little mixing of oxygenated and deoxygenated blood
- Two almost functionally separate circulations

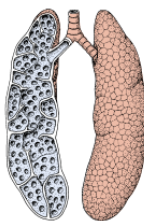


Class: Reptilia

Characteristics of Reptiles

Respiration

- Comparatively well developed lungs
- Depend exclusively on lungs for gas exchange
- Lungs are subdivided into numerous interconnecting air sacs that are highly vascularized
- Air is taken into the lungs by enlarging the thoracic cavity (Negative pressure mechanism)

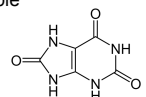
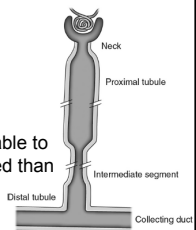


Class: Reptilia

Characteristics of Reptiles

Excretion

- Evolved efficient strategies for water conservation
- Nephrons lack the loop of Henle - unable to produce liquid urine more concentrated than their body fluid
- Nitrogenous waste - uric acid, water insoluble
- Salt glands (located near the nose or eyes) – secrete excess salts



Class: Reptilia

Characteristics of Reptiles

Movement on Land

- Have good body support and more efficiently designed limbs to travel on land
- Some have secondarily lost limbs

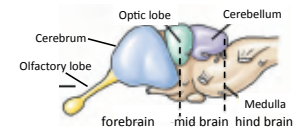


Class: Reptilia

Characteristics of Reptiles

Nervous & Sensory system

- Considerably complex, three regions: forebrain, mid brain, hind brain
- Cerebrum larger than in amphibians
- Crocodilians have a true cerebral cortex – allows complex behaviour
- Cerebellum- controls equilibrium, posture and movement
- Medulla - controls heartbeat, respiration, vascular tone, gastric secretions, and swallowing

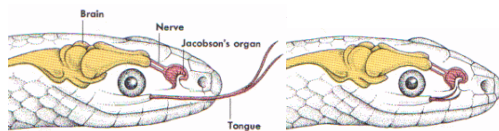


Class: Reptilia

Characteristics of Reptiles

Nervous & Sensory system

- Eyes - retinas have cone and rod cells
- Lizards – hearing not significantly developed (exception geckoes)
- Jacobson's organ - specialized olfactory chamber in the buccal cavity

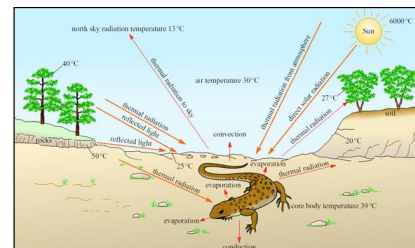


Class: Reptilia

Characteristics of Reptiles

Thermoregulation

- Ectothermic
- Behavioural thermoregulation

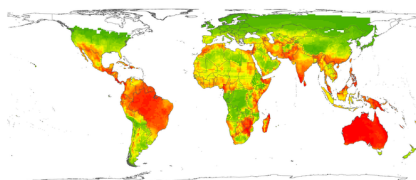


Class: Reptilia

Characteristics of Reptiles

Distribution

Distribution of global reptilian diversity: ~ 11,570 species
(Reptile database: <http://reptile-database.reptarium.cz/2022>)



Lizard species richness



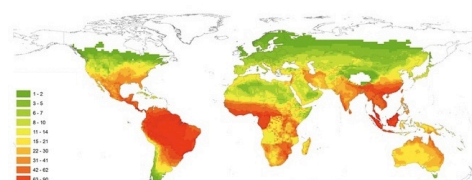
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Class: Reptilia

Characteristics of Reptiles

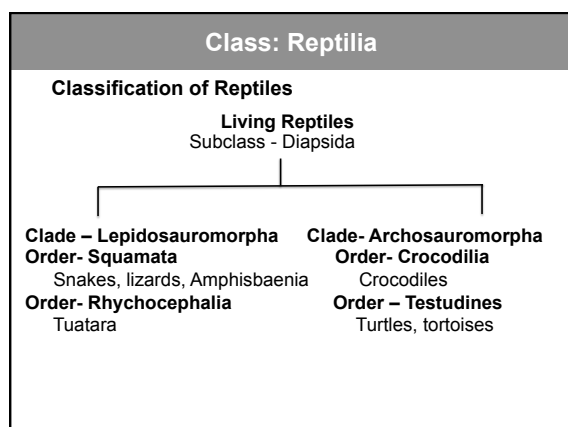
Distribution

Distribution of global reptilian diversity: ~ 11,570 species
(Reptile database: <http://reptile-database.reptarium.cz/2022>)



Snake species richness





Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

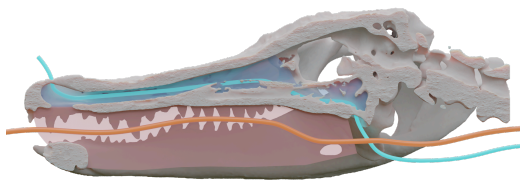
- Arose in the Cretaceous period (145 - 65 Mya)
- Have remained mostly unchanged for nearly 150 My
- Have an elongate and strong skull
- Massive jaw musculature - provide a wide gape and rapid, powerful closure
- Teeth are set in sockets – **Thecodont**




Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

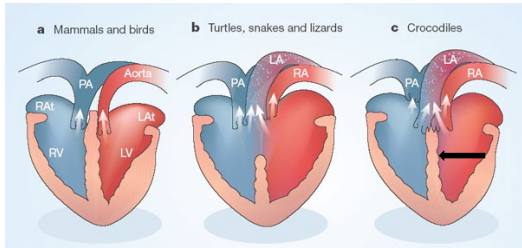
- Complete secondary palate present in the mouth– allows to breathe when the mouth is filled with water or food



Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Four chambered heart



Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Size – 1.5 m to 6 m, weighing up to 1000 kg
- Scutes in the skin are supported by bony plates - osteoderms




Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

- Reproduction: Oviparous, usually 20 to 50 eggs are laid in a mass of dead vegetation and guarded by the mother
- Temperature-dependent sex determination (TSD)
- Found in fresh and brackish water habitats in the tropical and subtropical regions in the world





Class: Reptilia

Order- Crocodilia: Crocodiles & Alligators

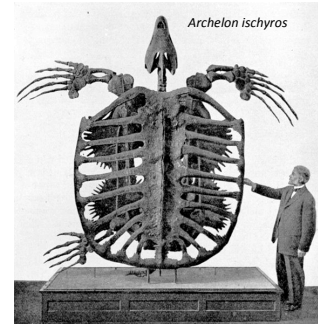
- Includes Alligators, Crocodiles, Caimans, Gharial



Class: Reptilia

Order- Testudines: Turtles & Tortoises

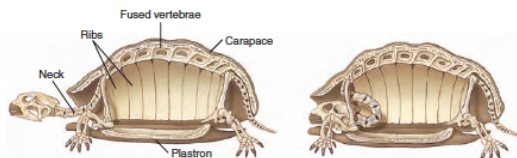
- Appear in the fossil record from the late Triassic, ~200 Mya
- Very little change in morphology since then



Class: Reptilia

Order- Testudines: Turtles & Tortoises

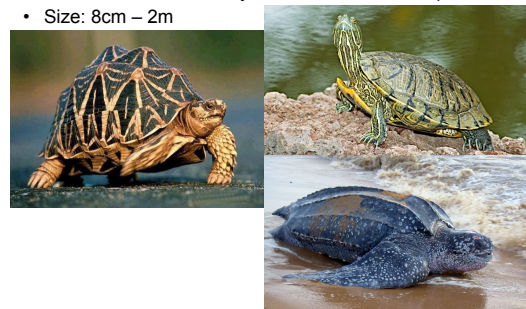
- Bodies are enclosed in shells consisting of a dorsal **carapace** and a ventral **plastron**
- Shell composed of two layers: outer horny layer of keratin
inner layer of bone
- Bony layer- fusion of ribs, vertebrae



Class: Reptilia

Order- Testudines: Turtles & Tortoises

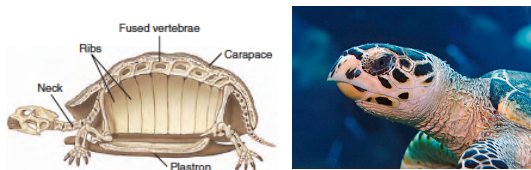
- Includes the modern day turtles, tortoises, terrapins etc.
- Size: 8cm – 2m



Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Limbs and limb girdles are located inside the rib cage
- Lacks teeth, jaw is provided with tough, horny plates for gripping food



Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Rigid shell with fused ribs - cannot expand its chest to breathe
- Air is taken in by increasing abdominal cavity volume by contracting posterior abdominal muscles (active process)
- Exhalation accomplished by contracting the pectoral girdle muscles compressing the viscera and forcing air out of the lungs (also an active process)

Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Brain- small, does not exceed 1% of the body weight
- Both middle and an inner ear present, perception of sound is poor
- Have a good sense of smell, vision and color perception
- Slow moving
- Diet – herbivores, omnivores and carnivores

Class: Reptilia

Order- Testudines: Turtles & Tortoises

- Reproduction – oviparous, internal fertilization, bury their shelled eggs in the ground
- Temperature-dependent sex determination (TSD)
Low temperatures during incubation - produce males
High temperatures during incubation - produce females



Class: Reptilia

Order- Rhynchocephalia: Tuataras

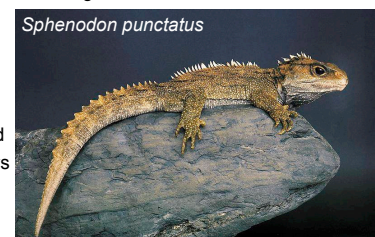
- Has Triassic origins (252-201 Mya)
- Has numerous features that are identical to 200 My old fossils
- Primitive diapsid skull – complete temporal bar



Class: Reptilia

Order- Rhynchocephalia: Tuataras

- Hemipenis not fully developed
- Lizard like body form, length 66 cm or less
- Represented by a single species
- Restricted to small islets in New Zealand
- Lives in burrows



Class: Reptilia

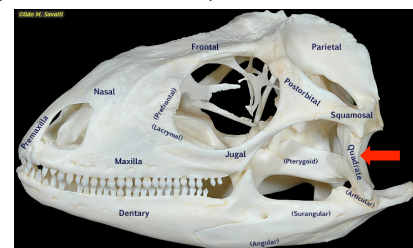
Order- Squamata: Lizards, Snakes & Worm Lizards

- Most recent and diverse of the diapsid evolution
- Includes ~95% of all known living reptiles
- Lizards appeared in Permian period (300 -250 MYA)
- Began their radiation in Cretaceous period (145-66 MYA)
- Snakes evolved from lizards during the late Cretaceous period
- Snakes evolved through extreme elongation of the body and displacement and rearrangement of internal organs
- Worm lizards, have structural specializations associated with burrowing

Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

- Body is covered with horny scales
- Squamates have movable quadrate bones



Class: Reptilia

Order- Squamata: Lizards, Snakes & Worm Lizards

Reproduction

- Shows both oviparity and ovoviviparity
- Ovoviviparity in reptiles is seen only among squamates
- Ovoviviparity evolved at least 100 separate times - associated with cold climates
- Ovoviviparity was achieved through increasing the length of time the eggs are kept within the oviduct
- Developing young respire through the chorion and obtain nutrition from yolk sacs

Class: Reptilia

Order- Squamata: Typical Lizards

- Extremely diverse group
- Terrestrial, burrowing, aquatic, and arboreal members
- Majority of lizards have four limbs and relatively short bodies
- In many, the limbs are degenerated or completely absent
- Movable eyelids
- External ear present
- Live in warm tropical and subtropical regions of the world
- Body temperature is kept relatively constant by behavioral thermoregulation

Class: Reptilia

Order- Squamata: Typical Lizards

Includes: Iguanas, monitor lizards, Chameleons, dragon lizards, geckoes, skinks



Dragon lizard



Gecko



Skink

Class: Reptilia

Order- Squamata: Typical Lizards

Includes: Iguanas, monitor lizards, Chameleons, dragon lizards, geckoes, skinks



Iguanas



Chameleon



Monitor lizards

Class: Reptilia

Order- Squamata: Worm Lizards

- Have elongate, cylindrical bodies
- Most lack external limbs
- Highly specialized burrowing forms
- Skin is soft and is divided into numerous rings
- Eyes and ears are hidden under skin - look like earthworms
- Distribution - South America and Southern North America and tropical Africa

Class: Reptilia

Order- Squamata: Worm Lizards



Class: Reptilia

Order- Squamata: Snakes

- Entirely limbless and lack both pectoral and pelvic girdles (exception pythons, boas)
- Vertebrae numerous shorter and wider than those of tetrapod reptiles – quick lateral undulations
- Ribs increase rigidity of the vertebral column



Class: Reptilia

Order- Squamata: Snakes

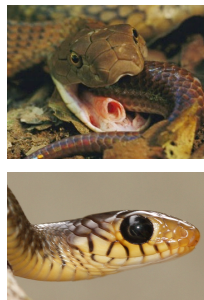
- Neural spine elevated - gives the muscles more leverage
- Highly **kinetic skull** – enable them to eat prey several times their own diameter
- Two halves of the lower jaw (mandibles) are joined only by muscles and skin
- Many of the skull bones are so loosely articulated (i.e. fitted)



Class: Reptilia

Order- Squamata: Snakes

- Tracheal opening (glottis) is found between the two mandibles
- Cornea of the eye is permanently protected with a transparent membrane – spectacle
- Fixed eyelids
- Vision is poor, arboreal snakes are an exception
- No external ears or tympanic membrane



Class: Reptilia

Order- Squamata: Snakes

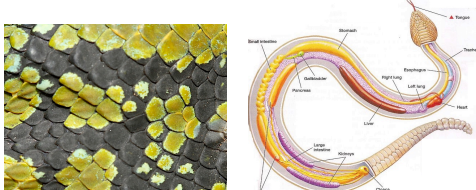
- Have internal ears (stapes present), hear low frequencies (100 to 700 Hz)
- Sensitive to vibrations carried in the ground
- Heavily depend on chemical senses - **Jacobson's organs**
- Some have heat-sensitive organs positioned in pits on their heads – pit organs (e.g. pit-vipers, pythons)



Class: Reptilia

Order- Squamata: Snakes

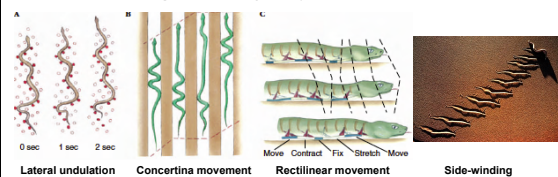
- Body is entirely covered with a tough, impervious skin
- Skin is covered with overlapping scales
- Extreme elongation - displacement and rearrangement of internal organs (one lung or left lung is reduced)



Class: Reptilia

Order- Squamata: Snakes

- Movement
 - Lateral undulation** - S-shaped path
 - Concertina movement** - S-shaped loops
 - Rectilinear movement** - straight line
 - Side-winding** – throwing body forward in loops



Class: Reptilia

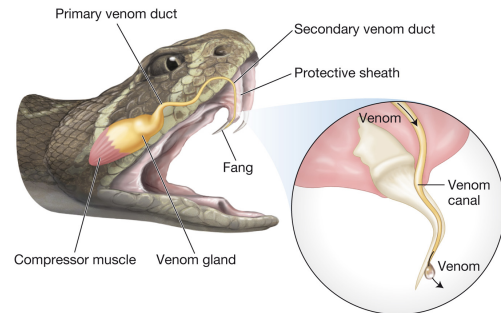
Order- Squamata: Snakes

- Some have a pair of teeth on the maxillary bones modified as fangs
- Fangs deliver venom that helps in immobilizing and digestion of prey
- Venom is stored in venom glands in the head, modified salivary glands
- Venom mixture of different types of,
 - a) Enzymes (90-95%)
 - neurotoxins (inhibits nerve transmission)
 - hydrolytic enzymes (digest cell membranes)
 - b) Polypeptides - Cytotoxins
 - c) Inorganic ions



Class: Reptilia

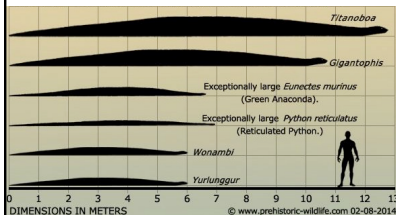
Order- Squamata: Snakes



Class: Reptilia

Order- Squamata: Snakes

Size – 10cm – 7m



Smallest snake in the world- Barbados thread snake (10 cm)



- Large extinct snakes – *Titanoboa*, *Gigantophis*, *Wonambi*, *Yurlunggur*

Class: Reptilia

Order- Squamata: Snakes



Class: Reptilia

Order- Squamata: Snakes

